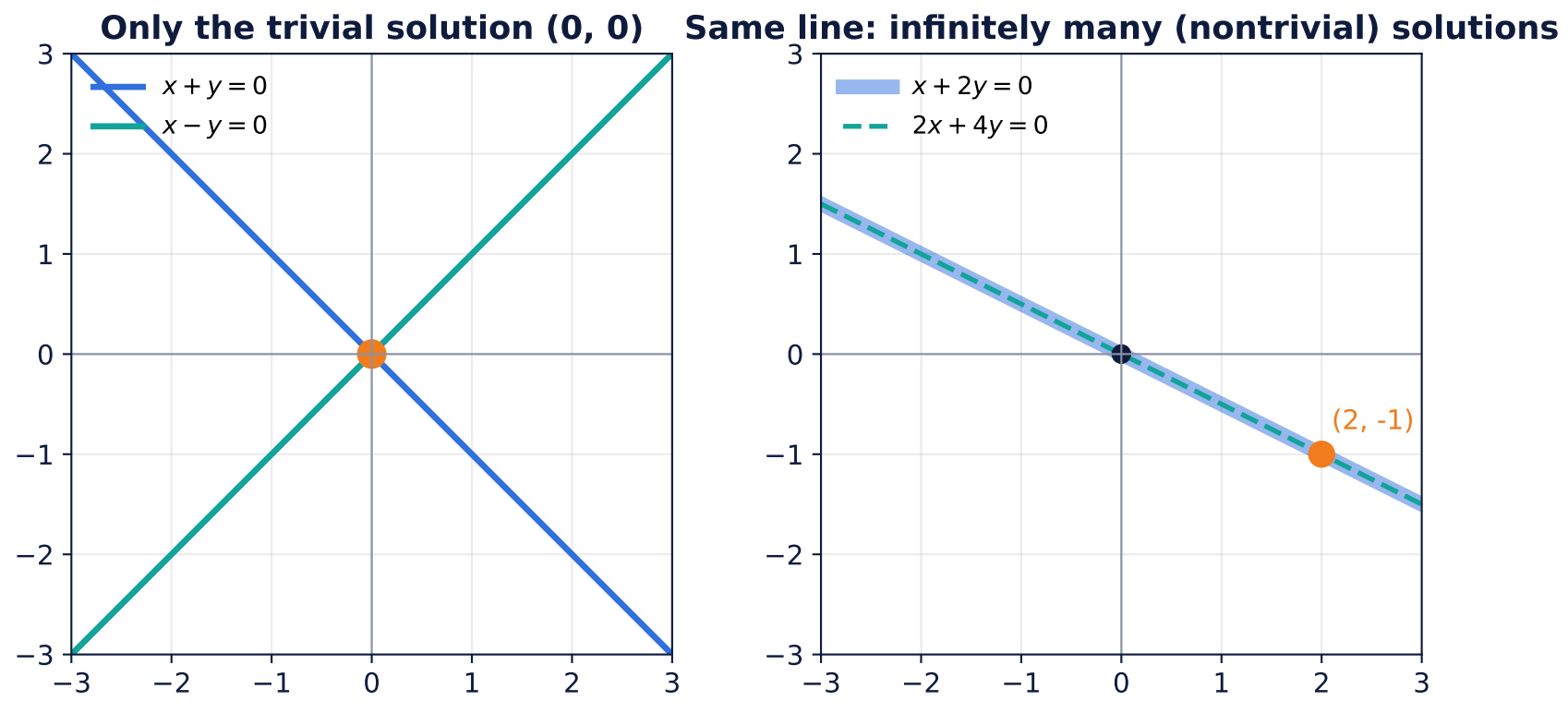
Only the trivial solution (0, 0)
$x + y = 0$
$x - y = 0$
Same line: infinitely many (nontrivial) solutions
$x + 2y = 0$
$2x + 4y = 0$
(2, -1)